

**ATTACHMENT 21 – INTERNAL MEMORANDUM FROM STAFF OF BECHTEL
/ PARSONS BRINCKERHOFF DATED AUGUST 31, 2000 RECOMMENDING
DISPOSITION OF DEFICIENCY REPORT NO. 1 BE CLOSED**
(12 pages)

MEMORANDUM

TO: Bob Steffy

FROM: Phil Aikele [REDACTED]

DATE: August 31, 2000

SUBJECT: Recommended disposition for DR #1

Subsequent to the writing of DR#1, all epoxy anchors in the subject area were retested at the higher load of 6,350 lb. There were a total of 19 anchors that failed out of the 187 that were retested at the higher load of 6,350lb vs. the initial contractly required load of 3,250. Refer to the break out below that gives the locations of the failures.

Module HOV078: 20 anchors retested, 3 failed. [located in mock-up] 15%
Module HOV079: 55 anchors retested, 14 failed [located in mock-up] 25%
Module HOV080: 56 anchors retested, 2 failed. 3.6%
Module HOV079: 56 anchors retested, 0 failed 0%

Each of the failed anchors was replaced and retested. Of these only one failed upon retesting and it was then replaced and retested again and passed. Hence, all existing anchors in the subject area have been tested and replaced as necessary and retested untill passing at the higher load. Copies of the test reports from Conam Inspection, Inc. are attached.

The "subject area" is comprised of all epoxy anchors that were installed by C09B2 in the ceiling of HOV in the C04A2 Contract area, which comprises the area where ceiling modules designated HOV078, HOV079, HOV080, HOV081 are located.

Observation: There were only two failures in the modules located outside of the ceiling mock-up area. This could be the result of better installation techniques being used by the contractor as gleaned from going through the "learning curve" in the mock-up area. These better installation techniques include: (1) cleaning holes out better and not using first epoxy out of the gun during the installation of the epoxy anchors and (2)not over torquing the nuts when the ceiling plates are installed over the anchors.

It is my recommendation that this DR#1 be closed.

Author: RESTEFFY at Cat01B

Date: 1/14/00 8:02 AM

Priority: Normal

TO: PCAIKELE (Phil Aikele), CFCARROL (Charles Carroll) at CAT01C, DEHATCH (David Hatch) at CAT01A,
AVSLAGLE (Ann Slagle) at CAT01C

Subject: DR-1 - Adhesive Anchors

----- Message Contents -----

Letter to MCC to revise A/B Deviation Report. Submittal for repairs is Code 3.
MCC to revise procedure and resubmit prior to removing damaged anchors.

The DR is for HOV ONLY Retesting at 6250# is required. Retest only HOV
including anchors that do not have ceiling installed. Time sheets will be used
to track costs.

The cost of repairs, etc will be an open issue, T&M sheets will document
actuals. Phil detailed inspections and monitoring will be required. Test data
will be required.

If retests prove good results EB/WB probably will not require retesting. If
results bad then a NEW Deviation will be issued. - Thanks - Bob

Conam Inspection
Inspection & Quality Services
2 Millbury Street
Auburn, Massachusetts 01501
Telephone = 1 508-832-5500
Facsimile = 1 508-832-5557



NONDESTRUCTIVE TESTING REPORT

CUSTOMER: Modern Continental Construction
ADDRESS: 6 Necco Ct
Boston MA 02110
ATTENTION: Jim Martin
LOCATION: Ted Williams Tunnel
JOB NO.: S8948 P.O. 9809-140642

INSPECTION METHOD

☐ VT ☐ MT ☐ PT ☐ UT ☐ RT ☐ ET ☒ OTHER

INSPECTION OF: Anchor resin to concrete – Tunnel Hoop Contract #C09B2

Line A Row 25 S1 – acceptable
Line A Row 29 S1 – acceptable
Line B Row 33 N1 – acceptable

INSPECTED BY:

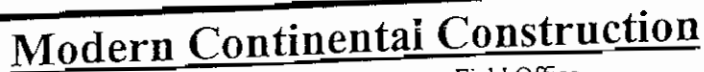

Lynn Anderson

LEVEL

II

DATE

8-23-00



Field Office
6 Necco Court
Boston, MA 02110
Phone: (617)695-2277
Fax: (617)695-2272

TRANSMITTAL
T0343

Central Artery/ Tunnel Project

99119-C09B2

I-90 Seaport Access Tunnel Finishes

PROJECT: I-90 Seaport Access Tunnel Finishes

DATE: 8/10/2000

TO: Bechtel/ Parsons Brinckerhoff
185 Kneeland Street
Boston, MA 02111

REF: H.O.V. Mock-up Anchor Pu
9809-1075-W

ATTN: Robert Steffy

RECEIVED
AUG 10 2000

WE ARE SENDING:		SUBMITTED FOR:	ACTION TAKEN:
☐ Shop Drawings	☐ Approval	☐ Approved as Submitted	
☐ Letter	☒ Your Use	☐ Approved as Noted	
☐ Prints	☐ As Requested	☐ Returned After Loan	
☐ Change Order	☐ Review and Comment	☐ Please Resubmit	
☐ Plans		☐ Submit	
☐ Samples	SENT VIA:	☐ For Your Information	
☐ Specifications	☒ Attached	☐ No Action	
☐ Other:	☐ Separate Cover Via		

ITEM NO.	COPIES	DESCRIPTION
----------	--------	-------------

SUBMITTAL	STATUS
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	
23	
24	
25	
26	
27	
28	
29	
30	
31	
32	
33	
34	
35	
36	
37	
38	
39	
40	
41	
42	
43	
44	
45	
46	
47	
48	
49	
50	
51	
52	
53	
54	
55	
56	
57	
58	
59	
60	
61	
62	
63	
64	
65	
66	
67	
68	
69	
70	
71	
72	
73	
74	
75	
76	
77	
78	
79	
80	
81	
82	
83	
84	
85	
86	
87	
88	
89	
90	
91	
92	
93	
94	
95	
96	
97	
98	
99	
100	

1	2	H.O.V. Mock-up Anchor Pull Test Results
---	---	---

Remarks:

ROUTING REQUEST			
99119 - C09B2			
A		W	WO
	DISTRIB.		
	RE	✓	
	ARE		✓
	OE	✓	
	AOE		
	LFE	✓	
	FE		
	C09B2-Sch	✓	
	C09B2 C C	✓	
	Proj Eng.	✓	
FILE	CN - 9 09B2		
FILE	CN - 9 09B2		
FILE	CN - 9 09B2		

Signed

~~Modern Continental~~



NONDESTRUCTIVE TESTING REPORT

CUSTOMER: Modern Continental Construction
ADDRESS: 6 Necco Ct
Boston MA 02110
ATTENTION: Jim Martin
LOCATION: Ted Williams Tunnel
JOB NO.: S8430 P.O. 9809-140642

INSPECTION METHOD

☐ VT ☐ MT ☐ PT ☐ UT ☐ RT ☐ ET ☒ OTHER

INSPECTION OF: Anchor resin to concrete – Tunnel Hov Contract #C09B2

Line B Row 25	N1	Fail		
Line B Row 26	N1	Fail	N2	Acceptable S1, S2 Fail
Line B Row 27	N1, S1	Acceptable		
Line B Row 28	N1	Acceptable	S1	Fail
Line B Row 29	N1, S1	Acceptable		
Line B Row 30	N1, S1	Fail		
Line B Row 31	N1, N2, S1, S2		Acceptable	
Line B Row 32	N1, S1	Acceptable		

Line C Row 25	N1, S1	Acceptable		
Line C Row 26	N1	Fail	S1	Acceptable
Line C Row 27	N1	Fail	S1	Acceptable
Line C Row 28	N1, N2	Fail	S1, S2	Acceptable
Line C Row 29	N1, S1	Acceptable		
Line C Row 30	N1	Fail	S1	Acceptable
Line C Row 31	N1, S1	Acceptable		
Line C Row 32	N1, S1	Acceptable		

INSPECTED BY: [REDACTED] LEVEL II DATE 7-11-00

Lynn Anderson



NONDESTRUCTIVE TESTING REPORT

CUSTOMER: Modern Continental Construction
ADDRESS: 6 Necco Ct
Boston MA 02110
ATTENTION: Jim Martin
LOCATION: Ted Williams Tunnel
JOB NO.: S8430 P.O. 9809-140642

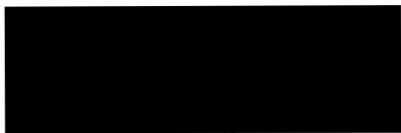
INSPECTION METHOD

☐ VT ☐ MT ☐ PT ☐ UT ☐ RT ☐ ET ☒ OTHER

INSPECTION OF: Anchor resin to concrete – Tunnel Hov Contract #C09B2

Line B Row 25 N1, S1 Acceptable
Line B Row 26 N1, S1, S2 Acceptable
Line B Row 28 S1 Acceptable
Line B Row 30 N1, S1 Acceptable

INSPECTED BY:



Lynn Anderson

LEVEL

II

DATE

7-13-00

Conam Inspection
Inspection & Quality Services
2 Milbury Street
Auburn, Massachusetts 01501
Telephone + 1 508-832-5500
Facsimile + 1 508-832-5557



NONDESTRUCTIVE TESTING REPORT

CUSTOMER: Modern Continental Construction
ADDRESS: 6 Necco Ct
Boston MA 02110
ATTENTION: Jim Martin
LOCATION: Ted Williams Tunnel
JOB NO.: S8430 P.O. 9809-140642

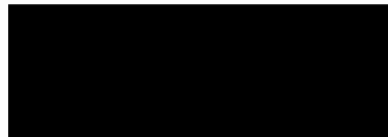
INSPECTION METHOD

☐ VT ☐ MT ☐ PT ☐ UT ☐ RT ☐ ET ☒ OTHER

INSPECTION OF: Anchor resin to concrete – Tunnel Hov. Contract #C09B2

Line C Row 26	N1	Acceptable
Line C Row 27	N1	Acceptable
Line C Row 28	N1, N2	Acceptable
Line C Row 30	N1	Acceptable

INSPECTED BY:



Lynn Anderson

LEVEL

II

DATE

7-13-00



NONDESTRUCTIVE TESTING REPORT

CUSTOMER: Modern Continental Construction
ADDRESS: 6 Necco Ct
Boston MA 02110
ATTENTION: Jim Martin
LOCATION: Ted Williams Tunnel
JOB NO.: S8503 P.O. 9809-140642

INSPECTION METHOD

☐ VT ☐ MT ☐ PT ☐ UT ☐ RT ☐ ET ☒ OTHER

INSPECTION OF: Anchor resin to concrete – Tunnel Flow Contract #C09B2

Line A Row 25	N1 – acceptable	S1 – Fail	REFLECTED 8/23 & PASSED
Line A Row 26	N1, S1 – acceptable		
Line A Row 27	N1, S1 – acceptable		
Line A Row 28	N1, N2, S1, S2 – acceptable		
Line A Row 29	N1 – acceptable	S1 – Fail	REFLECTED 8/23 & PASSED
Line A Row 30	N1, S1 – acceptable		
Line A Row 31	N1, S1 – acceptable		
Line A Row 32	N1, S1 – acceptable		

INSPECTED BY:



Lynn Anderson

LEVEL

II

DATE

7-17-00



NONDESTRUCTIVE TESTING REPORT

CUSTOMER: Modern Continental Construction
ADDRESS: 6 Necco Ct
Boston MA 02110
ATTENTION: Jim Martin
LOCATION: Ted Williams Tunnel
JOB NO.: S8503 P.O. 9809-140642

INSPECTION METHOD

☐ VT ☐ MT ☐ PT ☐ UT ☐ RT ☐ ET ☒ OTHER

INSPECTION OF: Anchor resin to concrete – Tunnel Box Contract #C09B2

Line A Row 33 N1, S1 acceptable
Line A Row 34 N1, S1 acceptable
Line A Row 35 N1, S1 acceptable

INSPECTED BY:



Lynn Anderson

LEVEL

II

DATE

7-19-00

Conam Inspection
Inspection & Quality Services

2 Millbury Street
Auburn, Massachusetts 01501
Telephone +1 508-832-5500
Facsimile +1 508-832-5557



NONDESTRUCTIVE TESTING REPORT

CUSTOMER: Modern Continental Construction
ADDRESS: 6 Necco Ct
Boston MA 02110
ATTENTION: Jim Martin
LOCATION: Ted Williams Tunnel
JOB NO.: S8503 P.O. 9809-140642

INSPECTION METHOD

☐ VT ☐ MT ☐ PT ☐ UT ☐ RT ☐ ET ☒ OTHER

INSPECTION OF: Anchor resin to concrete – Tunnel Hoop Contract #C09B2

Line B Row 33
Line B Row 34
Line B Row 35

✓ RETESTED 8/73 & PASSED
N1 – Fail, S1 – Acceptable
N1, N2, S1, S2 – Acceptable
N1 – Fail, S1 – Acceptable

Line C Row 33
Line C Row 34
Line C Row 35

RETESTED 8/3 & PASSED
N1, S1 – Acceptable
N1, S1 – Acceptable
N1, – Acceptable, S1 – Fail — RETESTED 8/3 & PASSED

INSPECTED BY:



Lynn Anderson

LEVEL

II

DATE

7-21-00



NONDESTRUCTIVE TESTING REPORT

CUSTOMER: Modern Continental Construction
ADDRESS: 6 Necco Ct
Boston MA 02110
ATTENTION: Jim Martin
LOCATION: Ted Williams Tunnel
JOB NO.: S8720 P.O. 9809-140642

INSPECTION METHOD

☐ VT ☐ MT ☐ PT ☐ UT ☐ RT ☐ ET ☒ OTHER

INSPECTION OF: Anchor resin to concrete - Tunnel (Hov) Contract #C09B2

Line B Row 33 N1 Fail *RETESTED 8/23 & PASSED*
Line B Row 35 N1 Acceptable

Line C Row 35 S1 Acceptable

Two (2) anchors Hov wall 1200 lbs Acceptable

Two (2) anchors bound wall 1200 lbs Acceptable

INSPECTED BY:



Lynn Anderson

LEVEL

II

DATE

8-3-00